



BRENT OZAR
UNLIMITED®

Cloud Deployment Methods

Note: this deck includes pricing & capacity slides.
I don't keep those up to date – they change constantly.

Deck 2 p1

Agenda: how do we buy this thing?

Choosing between PaaS and IaaS

Platform-as-a-Service
(Azure SQL DB, Amazon RDS)

Infrastructure-as-a-Service
(Amazon EC2, Azure VM, Google Compute Engine)



Deck 2 p2

Out of scope for this module

Comparing cloud vendors: you & I don't get to pick

Azure SQL DB Serverless

AWS Aurora and Babelfish

Azure SQL DB Hyperscale

Azure Synapse Analytics

Azure Arc

Deck 2 p3



Confusion- as-a-Service

Deck 2 p4



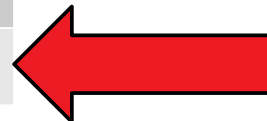
Who does what	Infrastructure as a service (Amazon EC2, Azure VMs, GCE)	Platform as a service (Amazon RDS, Azure SQL DB)
Who manages high availability?	You	Them
Who manages disaster recovery?	You	Them
Who manages patching?	You	Them
Who manages backups?	You	Them
Who can access backups?	You	MS: Them AWS: You



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Management	Infrastructure as a service (Amazon EC2, Azure VMs, GCE)	Platform as a service (Amazon RDS, Azure SQL DB)
Database features available	All	Most
Will you need a production DBA?	Yes	Maybe
Can just restore your databases to get them up into the cloud	Yes	MS: Depends AWS: Yes
Can just back up your databases to get them out of the cloud	Yes	MS: No AWS: Yes



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Infrastructure-as-a-service is kinda like sysadmin-as-a-service.



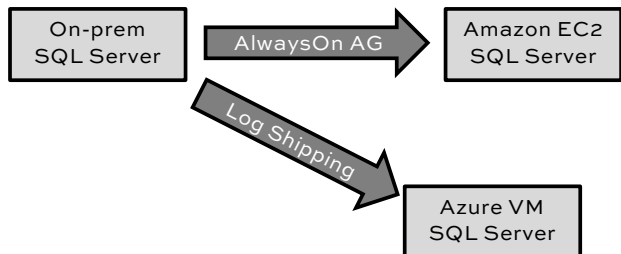
Deck 2 p7



Infrastructure-as-a-service is kinda like sysadmin-as-a-service.



Sure, we can build anything together.



Deck 2 p8



Platform-as-a-service is kinda like DBA-as-a-service.



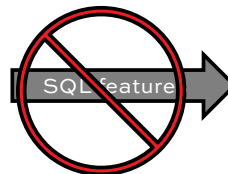
Deck 2 p9



Platform-as-a-service is kinda like DBA-as-a-service.



Your data in PaaS



Your servers anywhere else

Deck 2 p10



IaaS: you build and manage it.

It's just empty VMs.

Can use any version/edition of anything.

Performance: you can go as fast as your wallet is big.

Everything's negotiable. If you want to run your own custom version of something, or do something incredibly dumb, no one is going to tell you no.

Deck 2 p11



PaaS: someone else runs it.

Someone else handles the database administration, and there's no negotiations.

Features: Amazon RDS and Microsoft Azure SQL DB both support most (but not all) SQL Server features.

Remember, there are absolutely no negotiations. You can't turn on a feature they don't support.

This can be really frustrating for teams that have never heard "no" before.

PaaS is all or nothing: it doesn't integrate with your HA/DR.

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Making the decision	Infrastructure as a service (Amazon EC2, Azure VMs, GCE)	Platform as a service (Amazon RDS, Azure SQL DB)
Can fail over back and forth from their cloud to your on-premises servers	Yes	No
Your primary database server is:	Anywhere	In their cloud
Your secondary replicas are:	Anywhere	In their cloud
Your app servers are:	In the same place as the database	In the same place as the database
Best for companies that specialize in	Systems administration	Development

Deck 2 p13



Alex the Architect: PaaS.

Alex doesn't want to manage data HA/DR.

Alex is building new applications, and can stick to mainstream Azure SQL DB features.

In most cases – but not all – Alex is best off with Azure SQL DB (or groups of them.)

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Elliot's Enterprise: IaaS.

Already has large admin teams.

Used to dealing with complex servers.

Is usually under the gun to move to the cloud quickly, so does lift-and-shift.

Copies existing VMs up into the cloud.

Later, new apps or migrations might use PaaS – but might not even use SQL Server or Azure SQL DB for those.

Deck 2 p15



Billie's Small Business: mixed.



Billie often assumes a lift-and-shift is the only option.

But whenever individual apps & databases can be moved over to PaaS, they probably should to make Billie's life easier.

It's a matter of which apps support it.

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PaaS HA/DR:

Azure SQL DB, Amazon RDS

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The big players

Microsoft Azure SQL Database (one db at a time)

Microsoft Azure SQL DB Managed Instances
(like MS-managed Availability Groups)

Amazon Relational Database Service (RDS)
for SQL Server, MySQL, Postgres, Oracle

Google Cloud SQL for SQL Server, MySQL, Postgres

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First, look for showstoppers.

Whatever cloud vendor your company uses, provision their cheap PaaS database.

Deploy your existing application on there, and do a quick round of functionality testing.

Find out if your app relies on something a vendor doesn't support, because there's a LOT of things they don't support.

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Price is rarely the only factor.

Cloud vendors change their pricing constantly.

You'll also want to put your web and app servers up in the same cloud, so check pricing for those.

You may also rely on some of their related services already, like Amazon Redshift or Azure Synapse Analytics. In that case, be aware of bandwidth costs between vendor clouds.

Deck 2 p20



How Amazon & Google sell it

Per virtual server (like 2 cores 8GB RAM)

Can pay by the hour, or reserve instances for a year

Can rent your licenses from them, or bring your own
(but this gets really tricky:

<https://www.zdnet.com/article/microsoft-to-enact-new-cloud-outsourcing-and-hosting-licensing-changes-which-still-dont-address-core-customer-concerns/>

Deck 2 p21



EC2instances.info – click RDS tab

Instances

EC2

RDS

ElastiCache

Save 50% on EC2 with Autopilot, automated Reserved Instance purchasing.

Slack

Contribute

Region

US East (N. Virginia)

Pricing Unit

Instance

Cost

Hourly

Reserved

1 yr - No Upfront

Columns

Compare Selected

Clear Filters

Export

Search...

Name	API Name	Memory	Storage	vCPUs	SQL Server Standard Reserved cost
Filter..	Filter..	32	Min Storage: 0	8	Filter..
M6G Double Extra Large	db.m6g.2xlarge	32 GiB	0 GiB (EBS only)	8 vCPUs	unavailable
T2 General Purpose Double Extra Large	db.t2.2xlarge	32 GiB	0 GiB (EBS only)	8 vCPUs	unavailable
T4G Double Extra Large	db.t4g.2xlarge	32 GiB	0 GiB (EBS only)	8 vCPUs	unavailable
M6I Double Extra Large	db.m6i.2xlarge	32 GiB	0 GiB (EBS only)	8 vCPUs	\$18220.800000 annually
M5 Double Extra Large	db.m5.2xlarge	32 GiB	0 GiB (EBS only)	8 vCPUs	\$18220.800000 annually
M4 General Purpose Double Extra Large	db.m4.2xlarge	32 GiB	0 GiB (EBS only)	8 vCPUs	\$18203.280000 annually
M5D Double Extra Large	db.m5d.2xlarge	32 GiB	1 x 300 NVMe SSD	8 vCPUs	\$17804.700000 annually
T3 Double Extra Large	db.t3.2xlarge	32 GiB	0 GiB (EBS only)	8 vCPUs	\$17285.232000 annually

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Deck 2 p22



What it looks like to deploy

Choose a database creation method [Info](#)

☒ **Standard create**
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ **Easy create**
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [Info](#)

☐ Amazon Aurora


☐ MySQL


☐ MariaDB


☐ PostgreSQL


☐ Oracle


☒ Microsoft SQL Server


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Database management type [Info](#)

☒ **Amazon RDS**
RDS fully manages your database, including automatic patching. Choose this option if you don't need to customize your environment.

☐ **Amazon RDS Custom**
RDS manages your database and gives you privileged access to the OS. Use this option if you want to customize the database, OS, and infrastructure.

Edition

☐ **SQL Server Express Edition**
Affordable database management system that supports database sizes up to 10 GB.

☐ **SQL Server Web Edition**
In accordance with Microsoft's licensing policies, it can only be used to support public and Internet-accessible webpages, websites, web applications, and web services.

☐ **SQL Server Standard Edition**
Core data management and business intelligence capabilities for mission-critical applications and mixed workloads.

☒ **SQL Server Enterprise Edition**
Comprehensive high-end capabilities for mission-critical applications with demanding database workloads and business intelligence requirements.

License
license-included

Version
SQL Server 2019 15.00.4198.2.v1 ▼

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Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

- ☐ Standard classes (includes m classes)
- ☒ Memory optimized classes (includes r and x classes)
- ☐ Burstable classes (includes t classes)

db.r5.xlarge
4 vCPUs 32 GiB RAM Network: 4,750 Mbps



db.r5.xlarge
4 vCPUs 32 GiB RAM Network: 4,750 Mbps

db.r5.2xlarge
8 vCPUs 64 GiB RAM Network: 4,750 Mbps

db.r5.4xlarge
16 vCPUs 128 GiB RAM Network: 4,750 Mbps

db.r5.8xlarge
32 vCPUs 256 GiB RAM Network: 6,800 Mbps

db.r5.12xlarge
48 vCPUs 384 GiB RAM Network: 9,500 Mbps

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Storage

Storage type [Info](#)

Provisioned IOPS SSD (io1)
Flexibility in provisioning I/O

Allocated storage

20

GiB

Minimum: 20 GiB. Maximum: 16,384 GiB

Provisioned IOPS [Info](#)

1000

IOPS

Minimum: 1,000 IOPS. Maximum: 64,000 IOPS



Your actual realized IOPS might vary based on your database workload and DB Instance Class. Refer to the [here](#) rates section for more details.

Storage autoscaling [Info](#)

Provides dynamic scaling support for your database's storage based on your application's needs.

☒ Enable storage autoscaling

Enabling this feature will allow the storage to increase after the specified threshold is exceeded.

Maximum storage threshold [Info](#)

Charges will apply when your database autoscales to the specified threshold

1000

GiB

Minimum: 22 GiB. Maximum: 16,384 GiB

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Some parts are awesomely simple

Availability & durability

Multi-AZ deployment [Info](#)
☒ Yes (Mirroring / Always On)
☐ No

Multi-AZ deployment ×

Select Create Replica in Different Zone to have Amazon RDS maintain a synchronous standby replica in a different Availability Zone than the DB instance. Amazon RDS will automatically fail over to the standby in the case of a planned or unplanned outage of the primary.
[Learn More](#)

That's it. That's all there is.

You don't even pick which one it is.

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Some stuff isn't even shown by default.

► Additional configuration
Database options, encryption turned on, backup turned on, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned on.

 You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

Cancel

Create database

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Backup

- ☒ Enable automated backups
Creates a point-in-time snapshot of your database

Backup retention period [Info](#)

The number of days (1-35) for which automatic backups are kept.

▼ days

Backup window [Info](#)

The daily time range (in UTC) during which RDS takes automated backups.

- ☐ Choose a window
☒ No preference

- ☒ Copy tags to snapshots

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Alex, Billie, and Elliot love this.

Provisioning is really simple.

The hardware sizes are something familiar.

You get a lot of fine-grained control.

You don't get *complete* control –
but you get an awful lot.

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How Azure sells hardware

Azure SQL DB, one database:

- vCPUs (like 2 cores, 4 cores, 8, etc)
- Or Database Throughput Units (DTUs), a made-up performance number

Azure SQL DB groups: Elastic Pools of capacity

Azure SQL DB Managed Instances: like AGs, sold by server size (2 cpu, 4 cpu, etc)

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Microsoft Azure Search resources, services, and docs (G+)

Home > SQL databases >

Create SQL Database

Microsoft

Basics Networking Security Additional settings Tags Review + create

Create a SQL database with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize. [Learn more](#)

Did you know that new users in Azure can create a free Azure SQL Database and use it for 12 months using Azure free account? [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * SQL ConstantCare

Resource group * TrainingDemos2018 [Create new](#)

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name *

Server * firstresponderkit (West US 2)

Cost summary

General Purpose (GP_Gen5_2)	
Cost per vCore (in USD)	184.09
vCores selected	x 2
Cost per GB (in USD)	0.12
Max storage selected (in GB)	x 41.6
ESTIMATED COST / MONTH	372.97 USD

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Want to use SQL elastic pool? ? ☐ Yes ☒ No

Workload environment ☐ Development ☒ Production

i Default settings provided for Production workloads. Configurations can be modified as needed.

Compute + storage * ?

General Purpose
Standard-series (Gen5), 2 vCores, 32 GB storage, zone redundant disabled
[Configure database](#)

Backup storage redundancy
Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

Backup storage redundancy ? ☐ Locally-redundant backup storage ☐ Zone-redundant backup storage ☒ Geo-redundant backup storage

w Selected value for backup storage redundancy is Geo-redundant backup storage. Database backups will be geo-replicated which might impact your data residency requirements. [Learn more](#)

Deck 2 p33 Review + create Next : Networking >

This is where people struggle.

Service and compute tier

Select from the available tiers based on the needs of your workload. The vCore model provides a wide range of configuration controls and offers Hyperscale and Serverless to automatically scale your database based on your workload needs. Alternately, the DTU model provides set price/performance packages to choose from for easy configuration. [Learn more](#)

Service tier General Purpose (Scalable compute and storage options) v

Compute tier

Compute Hardware

Select the hardware configuration based on confidential computing hardware depends on

Hardware Configuration

vCore-based purchasing model

- General Purpose (Scalable compute and storage options)
- Hyperscale (On-demand scalable storage)
- Business Critical (High transaction rate and high resiliency)

DTU-based purchasing model

- Basic (For less demanding workloads)
- Standard (For workloads with typical performance requirements)
- Premium (For IO-intensive workloads)

up to 80 vCores, up to 415.23 GB memory
[Change configuration](#)

Deck 2 p34



So much more complex than SQL.

Here's what I recommend:
if you're migrating existing
production SQL Servers...

You probably want
Business Critical.

You probably want
vCore-based purchasing because that's what you're
used to measuring, and it's easier to migrate licenses.

General Purpose (Scalable compute and storage options)

vCore-based purchasing model

General Purpose (Scalable compute and storage options)

Hyperscale (On-demand scalable storage)

Business Critical (High transaction rate and high resiliency)

DTU-based purchasing model

Basic (For less demanding workloads)

Standard (For workloads with typical performance requirements)

Premium (For IO-intensive workloads)

Deck 2 p35



How Azure sells licensing

Azure SQL DB: licensing included in hourly/annual

Managed Instances: can bring your own licensing
from your Enterprise Agreement

I'm not going near any licensing discussions.

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How I handle licensing with clients

“Based on your workloads and HA/DR goals, here’s the cloud infrastructure we need.”

“Don’t make any long-term decisions yet: keep things flexible for the first 6 months, because our capacity planning might be off.”

“After 6-12 months, let’s look at things like:

- Reserved instances/capacity
- Elastic pools
- Licensing negotiations”

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Pains of PaaS

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Getting a copy of the database

“I want a copy of the production data on my machine”

AWS RDS: doable by writing a BAK file to S3

MS: not doable yet without doing an export, might be doable with Managed Instances & SQL Server 2022

Instead, switch to a development server in the cloud, and use the PaaS vendor’s backup/recovery system to present a copy of production to the dev server.

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Someone else manages versions

Amazon RDS supports several versions and editions of Microsoft SQL Server. The most recent supported version of each major version is listed here. For the full list of supported versions, editions, and RDS engine versions, see [Microsoft SQL Server versions on Amazon RDS](#).

- SQL Server 2019 CU15 15.00.4198.2, released per [KB5008996](#) on January 27, 2022.
- SQL Server 2017 CU27 14.00.3421.10, released per [KB5006944](#) on October 27, 2021.
- SQL Server 2016 SP3 13.00.6300.2, released per [KB5003279](#) on September 15, 2021.
- SQL Server 2014 SP3 CU4 SU 12.00.6433.1, released per [KB4583462](#) on January 12, 2021.
- SQL Server 2012: It's no longer possible to provision new instances in any Region. Amazon RDS is actively migrating existing instances off this version.
- SQL Server 2008: It's no longer possible to provision new instances in any Region. Amazon RDS is actively migrating existing instances off this version.

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Someone else manages features.

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_SQLServer.html#SQLServer.Concepts.General.FeatureNonSupport

In RDS, no FILESTREAM, FILETABLE, extended stored procedures, Machine Learning, R, log shipping, snapshots, replication, Resource Governor, server-level triggers, Service Broker, more.

Deck 2 p41



Someone else manages capacity

You can't design your own CPU/RAM/storage.

Amazon RDS limits the # of databases per server:

Instance class type	Single-AZ	Multi-AZ with DBM	Multi-AZ with Always On AGs
db.*.micro to db.*.medium	30	N/A	N/A
db.*.large	30	30	30
db.*.xlarge to db.*.16xlarge	100	50	75
db.*.24xlarge	100	50	100

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Platform-as-a-Service recap

The easy part:

they manage HA, DR, versions, patches, backups.

The hard parts:

- Hearing “no”
- Getting backups out to SQL Server
- Getting the sizing right in the beginning
(But who cares? That’s easy to adjust later.)

Any questions about PaaS?

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IaaS: VMs in AWS & Azure

Deck 2 p44



For the most part, it's just VMs.

Anything you can do on-premise, you can do here:

- AlwaysOn Availability Groups (with network gotchas)
- Database mirroring
- Log shipping, replication
- Reloading development servers
- Monitoring tools
- Painful troubleshooting

Deck 2 p45



But there are a few differences.

Deck 2 p46



Licensing by the host

SQL Server Standard Edition: pay by the VM guest

SQL Server Enterprise Edition:

- License the whole VM physical host
- Run as many VMs as you want on that host
- All VMs get Enterprise Edition features
- Common deployment method for enterprises

Deck 2 p47



You still can.

Get your very own box in the cloud where you can run your own SQL Server VMs.

License the host rather than each guest. Big savings.

Amazon EC2 Dedicated Hosts

An Amazon EC2 Dedicated Host is a physical server with EC2 instance capacity fully dedicated to your use. Dedicated Hosts can help you address compliance requirements and reduce costs by allowing you to use your existing server-bound software licenses. Visit the [EC2 Dedicated Host Pricing page](#) for information on availability and pricing.

Amazon EC2 Dedicated Hosts

Sign Up Now



Benefits



Save Money on Licensing Costs

Dedicated Hosts allow you to use your existing per-socket, per-core, or per-VM software licenses, including Microsoft Windows Server, Microsoft SQL Server, [SUSE Linux Enterprise Server](#), [Red Hat Enterprise Linux](#), or other software licenses that are bound to VMs, sockets, or physical cores, subject to your license terms. Learn more about your [Windows licensing options](#).



Help Meet Compliance and Regulatory Requirements

When you launch instances on a Dedicated Host, the instances run on a physical server that is dedicated for your use. While Dedicated instances also run on dedicated hardware, Dedicated Hosts provide further visibility and control by allowing you to place your instances on a specific, physical server. This enables you to deploy instances using configurations that help address corporate compliance and regulatory requirements.

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But the types of VMs are fixed.

Dedicated Host Attributes			Instance Capacity Per Host								
Instance Type	Sockets	Physical Cores	medium	large	xlarge	2xlarge	3xlarge	4xlarge	6xlarge	8xlarge	9xlarge
c3	2	20	-	16	8	4	-	2	-	1	-
c4	2	20	-	16	8	4	-	2	-	1	-
c5	2	36	-	36	18	8	-	4	-	-	2
c5n	2	36	-	36	18	8	-	4	-	-	2
c5d	2	36	-	36	18	8	-	4	-	-	2
p2	2	36	-	-	16	-	-	-	-	2	-
p3	2	36	-	-	-	8	-	-	-	2	-
p3dn	2	36	-	-	-	4	-	-	-	1	-

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Next, VMs are disposable.

Conceptually, think of your application as groups of databases that can live anywhere.

Don't think just different servers – think different cloud providers, thousands of miles apart.

(This is what happens when you fail over a server.)

Back up data, not servers.

One of my clients loses a VM every 29 days on avg.

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Vanishing ephemeral storage

AWS calls it Ephemeral, Azure calls it Temporary

Included with most instance types

Bigger instances = more space, volumes

Typically solid state

Great for TempDB, staging databases

Deck 2 p51



Sometimes for user databases!

If servers are disposable, why not put your databases on really fast ephemeral solid state storage?

If the VM dies, just fail over to another VM.

Problem: you have to re-initialize your failover mechanism by restoring your databases again from scratch.

You have to be *really* confident in your HA/DR.

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Example: new PCIe SSD storage

Very, very fast solid state on PCI Express cards

Included free with many instance types

Name	API Name	Memory	Instance Storage	Windows On Demand cost
Search	Search	Search	Search	Search
i3 High I/O Double Extra Large	i3.2xlarge	61.0 GiB	1900 GiB NVMe SSD	\$724.160000 monthly
i3EN Double Extra Large	i3en.2xlarge	64.0 GiB	5000 GiB (2 * 2500 GiB NVMe SSD)	\$928.560000 monthly
M2 High Memory Quadruple Extra Large	m2.4xlarge	68.4 GiB	1680 GiB (2 * 840 GiB HDD)	\$1007.400000 monthly
M5AD Quadruple Extra Large	m5ad.4xlarge	64.0 GiB	600 GiB (2 * 300 GiB NVMe SSD)	\$1138.800000 monthly
D2 Double Extra Large	d2.2xlarge	61.0 GiB	12000 GiB (6 * 2000 GiB HDD)	\$1168.730000 monthly
M5 General Purpose Quadruple Extra Large	m5d.4xlarge	64.0 GiB	600 GiB (2 * 300 GiB NVMe SSD)	\$1197.200000 monthly
Z1D 3xlarge	z1d.3xlarge	96.0 GiB	450 GiB NVMe SSD	\$1217.640000 monthly
H1 Quadruple Extra Large	h1.4xlarge	64.0 GiB	4000 GiB (2 * 2000 GiB HDD)	\$1220.560000 monthly
R5AD Quadruple Extra Large	r5ad.4xlarge	128.0 GiB	600 GiB (2 * 300 GiB NVMe SSD)	\$1302.320000 monthly
R5D Quadruple Extra Large	r5d.4xlarge	128.0 GiB	600 GiB (2 * 300 GiB NVMe SSD)	\$1378.240000 monthly

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Hardware migration

On-premise: if you want to upgrade a VM to faster processors, you just vMotion or LiveMigrate it.

In the cloud, you provision a new VM on the new instance type you want.

Instance types come out frequently.

The new ones are often faster/cheaper.

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Recap



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What we covered

Platform-as-a-Service (Azure SQL DB, Amazon RDS) is best for developer shops who don't want DBAs.

- Alex the Architect defaults to this.
- Billie's Business should use it wherever possible.

Infrastructure-as-a-Service (Amazon EC2, Azure VM) is like building blocks, and you have new techniques.

- Elliot's Enterprise prefers this.

Both keep changing really, really quickly.

